BROAD EXAMINATIONS®

P.7 MATHEMATICS EXAMINATION

PRE – PLE TRIAL SET I 2023

Time allowed: 2 hours 15 minutes.

Random No.						Personal No.		

Candidate's Name:.....

Candidate's Signature.....

District Name: .....

Read the following instructions carefully:

1. This paper is made up of two sections: **A** and **B**.
2. Section **A** has **20** questions (**40 Marks**)
3. Section **B** has **12** questions (**60 Marks**)
4. Answer **ALL** questions in both sections **A** and **B**.
5. All answers must be written in the space provided in blue or black ball point pens and ink. **Only diagrams should be done in pencil.**
6. Unnecessary crossing of answers will lead to loss of marks.
7. Any handwriting, which cannot be easily read, may lead to loss of marks.
8. Do **not** fill anything in the boxes indicated for Examiners' use only.

FOR EXAMINERS' USE ONLY

PAGES	MARKS	SIGN
Page 2		
Page 3		
Page 4		
Page 5		
Page 6		
Page 7		
Page 8		
TOTAL		

Teacher's comment to the learner

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Approved by:

Team Head Mathematics Department

PLE REVISION TIPS

In groups of fives, revise with your friends simple work from P.4 class.

Read about:

- 3- Dimensional figures and their properties
- Tally marks under data handling
- Addition and subtraction of time
- 4 by 4 magic squares and magic triangles

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SECTION .A. (40 Marks)

1. Add: 43 + 16

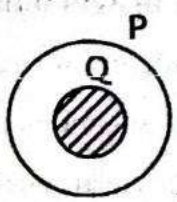
2. Change 13_{five} to base ten.

3. Work out the LCM of 12 and 15

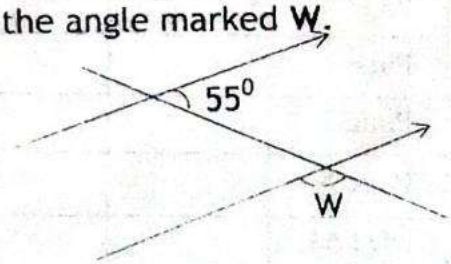
4. Find the range of 3, 7, 8, 5 and 6.

5. A trader sold a dozen picfare books at sh.1200 each book. How much money did the trader get after selling the books?

6. Describe the shaded region in the figure below.



7. In the diagram below, find the size of the angle marked W.



8. What number is increased by 10% to become 22?

9. A driver drove his car at a speed of 20 metres every second. What was his speed in km/h?

10. **Solve:** m + 3 = 1 (mod.7)

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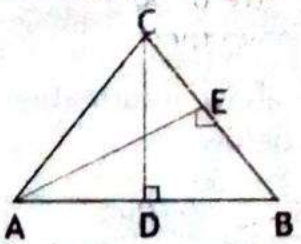
11. Simplify: $3x - (x + 4)$

12. In the diagram below, AB = 8cm, BC = 12cm and AE = 4cm.



Find the length DC.

13. Write the time 3:50pm in 24 hour clock system.

14. Round off 18.87 to the nearest whole number.

15. If y = x - 3, complete the table below.

x	3	0	____
y	6	____	4

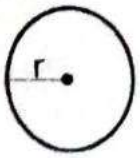
16. Using a ruler, a pencil and a pair of compasses only, construct an angle of 15^0

17. Express 14 as a product of its prime factors.

18. Set G = {2, 0, 5, 9, 7}. Find the number of proper subsets in set G.

19. Given that $\frac{2}{3}$ of a kilo of maize flour costs sh.1600, find the cost of 4kg of maize flour at the same rate.

20. The diagram below is a circular end of a cylindrical tin whose circumference is 88 metres. Find its radius.



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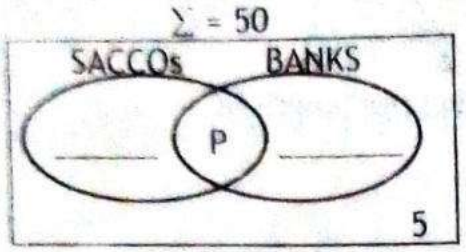
SECTION .B. (60 Marks)

21. In a trading centre of 50 traders, 32 have accounts in SACCOs (S), 34 have them in Banks (B), some (P) have them in both institutions while 5 do not have accounts at all.

Represent the above information in the venn diagram below.



(b) How many traders have accounts in both institutions?

(c) If a trader is selected at random to be a business representative, find the probability of choosing one who has an account in SACCOs only.

(05 Marks)

22. (a) **Work out:**

$$\begin{array}{r} 1\ 1\ 0_{two} \\ -\ 1\ 1_{two} \\ \hline \end{array}$$

(b) If $102p = 201_{five}$, find the unknown base p.

(05 Marks)

23. Study the exchange rates below and answer the questions that follow;

1 United States dollar (US$1) = Ugsh.3500
1 Kenya shilling (Ksh.1) = Ugsh.30
1 British pound Sterling (£.1) = Ugsh.4000

(a) A man has 330 US dollars and Ksh.5500. How much money in Uganda shillings does he have altogether?

(b) A phone costs Ugsh.640,000. Find its cost in pounds.

(05 Marks)

24 The number of pupils in P.1, P.2, P.3 and P.4 at Excel Junior School are 66, 54, 72 and 48 respectively.

Which class has the least number of pupils?	(b) How many more pupils are there in P.3 than in P.1?

(c) Work out the average number of pupils in the four classes.

(05 Marks)

25. (a) Using a ruler, a pencil and a pair of compasses only, construct a triangle PLE where line PL = 7cm, angle PLE = 75^0 and angle EPL = 45^0

Measure line LE.

(05 Marks)

26 (a) Work out. 1.36 - 2.9

(b) Simplify. 7 - 12.12 - 6.1

(05 Marks)

27 Today, Monday 14th September, teachers at Pole pole Primary School have received their salaries. They will receive their next salary after 30 days. On which day and date will they receive their next salary?

(04 Marks)

28 Solve;

(a) 3a - 5 = 19

(b) 5k - 3 = k + 5

(05 Marks)

29 A water tank is 80cm long, 60cm wide and 90cm high.

(a) Find its volume.

(b) How much water does it hold when it is $\frac{5}{9}$ full?

(05 Marks)

Sarah, Susan, Barbra and Monius have an average age of 22 years. Sarah is n years old, Susan is 5 years older than Sarah, Barbra is 19 years old and Monius is n+4 years old.

Find the value of n.

(b) Find Susan's age.

(05 Marks)

A motorist left town K at 8:30am at a speed of 35km/h. After a 2 hour drive, the car go a problem which took him 30 minutes to solve. He then continued to town L at a speed of 40km/h reaching town L at 12:15pm. He immediately returned to town K at a constar speed of 96km/h.

How far is town L from town K?

(b) Calculate the average speed of the motorist for the whole journey.

(05 Marks)

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The pictograph below shows the number of boxes of books received by UPE Schools in a Sub county.

SCHOOL	NO. OF BOXES.
Ntusi	⊠ ⊠ ⊠
Nzozi	⊠ ⊠ ▽
Busega	⊠ ⊠ ⊠ ⊠
Bitaabo	⊠ ⊠ ▽

Given that ⊠ represents 24 books and ▽ represents12 books,

Find the number of books Busega Primary School received.

(b) How many more books did Ntusi get than Bitaabo Primary School?

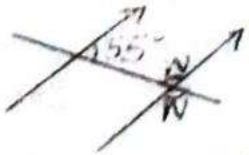
Work out the total number of books the four schools received.

END

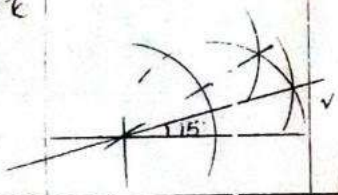
(06 Marks)

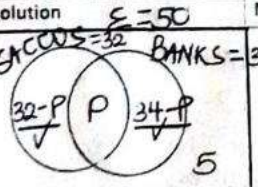
BROAD EXAMINATIONS GUIDES

P.7 MATHEMATICS PRE-PLE TRIAL SET I TERM III 2023

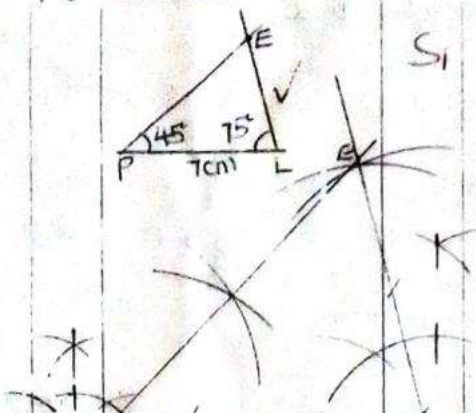
Qn.	Class	Solution	Marks	Comments
1	P1	43 +16 59 ✓	B2	on sight
2.	P5	(1×5) + (3×5) ✓ 1×5 + 3×1 5 + 3 8ten ✓	M1 A1	for expanding for 8ten
3	P4	2 | 12 | 15 2 | 6 | 15 3 | 3 | 15 5 | 1 | 5 | 1 | 1 2×2×3×5 4×15 60 ✓	M1 A1	Accept use of multiples M1 for correct method A1 for 60
4	P5	Range = H − L = 8 − 3 ✓ = 5 ✓	M1 A1	for subtracting for 5
5	P4	sh 1200 × 12 ✓ sh 14400 ✓	M1 A1	for multiplying for sh 144,000
6	P5	G ∩ P or set G ✓	B2	On sight
7	P.6	 W + 55° = 180° (co-int angles) W + 55° − 55° = 180° − 55° W = 125° A1	M1	for eqn for 125°

Qn.	Class	Solution	Mrks	Comments
8	P6	let the no be n 100% + 10% = 110% ✓ $\frac{110}{100} \times n = 22$ M1 $\frac{100}{10} \times \frac{4n}{100} = \frac{25 \times 110}{1}$ n = 20 ✓	A1	for correct method for 20
9.	P6	20 mts = $\frac{20\text{ m}}{1\text{ sec}}$ = $\frac{20\text{ km}}{1000} - \frac{3}{3600}$ hr ✓ = $\frac{20}{1000}$ km × $\frac{3600}{1\text{ hr}}$ = 72 km/hr ✓	M1 A1	for dividing for 72 km/h
10	P7	m + 3 = 1 (mod 7) m + 3 − 3 = 1 − 3 (mod 7) ✓ m = 1 − 3 (mod 7) m = 8 − 3 (mod 7) m = 5 (mod 7) ✓	M1 A1	for subtracting 3 for m = 5
11	P6	3x − (x + 4) 3x − x − 4 ✓ 2x − 4 ✓	M1 A1	for removing brackets
13.	P6	3 50 + 12 00 ✓ 15 50 hrs	M1 A1	for adding for 15 50 hrs

Qn	Class	Solution	Mrks	Comments
12	PE	$\frac{1}{2}bh = \frac{1}{2}bh$ $\frac{1}{2}\times\frac{3}{6}\times h = \frac{1}{2}\times 12\times\frac{3}{4}$ ✓ $\frac{2h}{2} = \frac{24}{2}$ $h = 12cm$ ✓	M1 A1	for correct method for 12cm
14	PE	TO th 18 ~~32~~ + 1 19 ✓	M1 A1	for adding 1 for 19
15	P7	Value of x $y = x-3$ $4 = x-3$ $4+3 = x-3+3$ $7 = x$ ✓ Value of y $y = x-3$ $y = 0-3$ $y = -3$ ✓	B1 B1	for any correct entry 7 and -3
16	PE	 ✓	B2	for a well constructed angle
17	PE	2 | 14 7 | 7 ✓ 1 $14 = 2\times 7$ ✓	M1 A1	for correct method for 2x7
18	PE	2^5-1 2^5-1 ✓ M1 $2\times2\times2\times2\times2-1$ $32-1$ 31 proper subsets	A1	for 2^5-1 for 31
19	PE	$\frac{2}{3}$ kg → sh 1600 1 kg → sh 1600 ÷ $\frac{2}{3}$ → sh ~~1600~~ 2400 × $\frac{3}{2}$ → sh 2400 4kg → sh 2400 ✓ × 4 sh 9600 ✓	M1 A1	for multiplying for sh 9600
20	PE	$2\pi r = C$ $2\times\frac{22}{7}r = 88$ ✓ M1 $\frac{7}{44}\times\frac{44}{7}r = \frac{88}{2}\times\frac{7}{44}$ $r = 14$ metres ✓	A1	for correct substitution for 14m
		or $\pi d = C$ $7\times\frac{22}{7}d = 88\times7$ ✓ $\frac{22d}{22} = \frac{88\times7}{22}$ $d = 28$ radius = 14m ✓	M1 A1	for correct method for correct

Qn.	Class	Solution	Mrks	Comments
21	P6 (a)	$\varepsilon = 50$ SACCOS = 32 BANKS = 34  5	B1 B1	for 32-P for 34-P
	(b)	(32+34+5) − 50 ✓ 71−50 21 traders ✓	M1 A1	for a correct method for 11
	(c)	n(S) only 32−21 11 ✓ Prob = $\frac{11}{50}$ ✓	B1	for $\frac{21}{50}$
22 (a)	P7	0 8 2 1 1 0 two − 11 two 11 two ✓ B2	05	on sight
(b)		$102_p = 201_{five}$ $(1\times P^2)+(2\times P^0) = (2\times5^2)+(1\times5^0)$ ✓ M1 for expanding $P^2+(2\times1) = 2\times5\times5+(1\times1)$ $P^2+2 = 50+1$ $P^2+2-2 = 51-2$ ✓ M1 for subtracting 2 $\sqrt{P^2} = \sqrt{49}$ $P = 7$ P = seven ✓ A1 for 7	05	

Qn.	Class	Solution	Mrks	Comments
23	P6	(a) US dollars to Ugsh 330 × Ugsh 3500 Ugsh 1155000 ✓ Ksh to Ugsh 5500 × Ugsh 30 Ugsh 165000 ✓ TOTAL Ugsh 1155000 + Ugsh 165000 Ugsh 1320000 ✓	B1 B1 B1	for Ugsh 1155000 for Ugsh 165000 for Ugsh 1320000
		b) ~~Ugsh 640,000~~ 160 ~~Ugsh 4000~~ ✓ 05 160 pounds ✓	M1 A1	for dividing for 160
24	P5	(a) P4 ✓	B1	
		b) ~~72~~ 6 −66 6 more pupils	M1 A1	for subtracting for 6
		c) 66+54+72+48 ✓ 4 ~~240~~ 60 ~~4~~ 1 60 pupils ✓	M1 A1	for adding and dividing for 60
			05	

Qn.	Class	Solution	Mrks	Comments
25	P6	Sketch 	S_1	in a detailed sketch
			C_1	for 45°
			C_1	for 75°
			L_1	for 7cm
		(b) LE = 5.9 ± 1 cm ✓	B5	K_1
26	P5	(a) 4.86 − 2.9 = 1.46 ✓	M1	for arranging according to PV
			A1	for 1.46
		(b) (7 + 6.3) − 12.12	B1	for 13.3
		7.0 + 6.3 = 13.3 ✓; 13.30 − 12.12 = 1.18 ✓	M1	for subtracting
			A1	for 1.18
			05	

Qn.	Class	Solution	Mrks	Comments
27	P7	1 + 30 = ___ (mod 7) ✓	M_1	
		31 = ___ (mod 7)		
		31 − 7 = 4 r 3		
		31 = 3 (mod 7) ✓	A_1	
		The day will be Wednesday	✓B_1	for Wednesday
		Days remaining in Sept. 30 − 14 = 16		
		Days to cover in Oct. 30 − 16 = 14		
		The date will be Wednesday, 14th October ✓	B_1	for correct date
			04	
28	P6	(a) 3a − 5 + 5 = 19 + 5 ✓	M_1	for collecting like terms
		$\frac{3a}{3} = \frac{24}{3}$		
		a = 8 ✓	A_1	for [illegible]
		(b) 5k − 3 = k + 5		
		5k − k = 5 + 3 ✓	M_1	for [illegible] like terms
		$\frac{4k}{4} = \frac{8}{4}$ ✓	M_1	for [illegible]
		k = 2 ✓	A_1	for k = 2
			05	

Qn.	Class	Solution	Mrks	Comments
29	P6	(a) 80 cm × 60 cm × 90 cm ✓	M_1	emphasize correct units
		4800 cm² × 90 cm		
		432000 cm³ ✓	A_1	
		(b) $\frac{5}{9}$ × 4320 = 2400 cm³		
		5 × 4800 cm³		
		24000 cm³ ✓	B_1	
		Water in litres $\frac{24000\,cm^3}{1000\,cm^3}$ ✓	M_1	
		24 litres ✓	A_1	
30	P6	(a) Total age = 22 × 4 = 88 yrs ✓	B_1	for sum
		Sarah — n		
		Susan — n + 5		
		Nenius — n + 4		
		Barbra — n + 4 + 19 = n + 23		
		n + n + 5 + n + 4 + n + 23 = 88 ✓	M_1	for eqn
		n + n + n + n + 9 + 23 = 88		
		4n + 32 − 32 = 88 − 32 ✓	M_1	
		$\frac{4n}{4} = \frac{56}{4}$		
		n = 14 ✓	A_1	for 14
		(b) Susan → 14 + 5 = 14 + 5 = 19 yrs ✓	B_1	

Qn.	Class	Solution	Mrks	Comments
31	P7	(a) Dist covered before problem $\frac{35\,km}{1\,hr} \times 2\,hr$ = 70 km ✓	B_1	for 70 km
		Total time from K to L 12.15 − 8.30 = 3.45 = $3\frac{3}{4}$ hrs		for [illegible] time
		Time taken after the problem $3\frac{3}{4} - (2\,hrs + \frac{1}{2}\,hr)$		
		$3\frac{3}{4} - 2\frac{1}{2}$		
		3.45 − 2.30 = 1.15 ✓ = $1\frac{1}{4}$ hrs ✓		
		Distance covered after rest $40 \times 1\frac{1}{4}$		
		$40 \times \frac{5}{4}$ = 50 km ✓	B_1	for [illegible]
		Total distance 70 km + 50 km = 120 km ✓	B_1	for [illegible]

Qn.	Class	Solution	Mrks	Comments	Qn.	Class	Solution	Mrks	Comments
31	P7	(b) Total distance covered 120 km + 120 km = 240 km ✓ Time taken returning $\frac{\cancel{120}^{10}}{\cancel{96}_{8}}$ hrs $\frac{\cancel{10}^{5}}{\cancel{8}_{4}}$ hrs $1\frac{1}{4}$ hrs ✓	B1	for $1\frac{1}{4}$ hrs					
		Total time taken $3\frac{3}{4} + 1\frac{1}{4} = 3 + 1 + \frac{3}{4} + \frac{1}{4}$ $= 4 + 1$ $= 5$ hrs A S = $\frac{\cancel{240}^{48}\text{km}}{\cancel{5}_{1}\text{hrs}}$							
	(05)	= 48 km/hr ✓	B1	for 48 km/hr					
32	P5	a) 24 × 4 ✓ = 96 ✓ books	M1 A1	for multiplying or adding					
		b) $3 - 2\frac{1}{2} = \frac{1}{2}$ $\frac{1}{\cancel{2}_{1}} \times \cancel{24}^{12}$ ✓ = 12 more books	M1 A1	for multiplying					
		c) 24 × 12 ✓ 288 books	M1 A1	for multiplying					